

RESERVOIR ENGINEERING — EXAMPLE STUDY

Simulation Study

BLACK_OIL_DEMO Field: History Match & 10-Year Production Forecast

Full-cycle reservoir simulation study — static model review, fluid characterisation, assisted history matching and decline-curve forecasting — executed by an AI agent connected to **tNavigator** through the Local MCP Server, under engineering supervision.

215,040	15	616	65.1 → 2.77	2.46 M sm ³
GRID CELLS (48×35×128)	WELLS (9 PROD / 6 INJ)	SIMULATIONS RUN	OBJECTIVE FUNCTION	FIELD EUR

PREPARED BY Rock Flow Dynamics tNavigator 26.2 + AI agent	PREPARED FOR TIMOR GAP, E.P. capability demonstration	DATE August 2026 Perth, Australia
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About this example. Prepared with tNavigator 26.2 to demonstrate AI-agent reporting in a TIMOR GAP-style template. All results derive from the synthetic BLACK_OIL_DEMO tutorial dataset; this document is not an official TIMOR GAP publication and contains no TIMOR GAP data.

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About this report. A complete simulation study of the BLACK_OIL_DEMO field model: the static model and grid, reservoir properties, fluid and rock physics, production history, assisted history matching with sensitivity screening, and a 10-year decline-curve forecast cross-checked against the matched model. Every figure is generated from the actual study data. The workflow — 616 simulation runs, diagnostics, dashboards and this document — was executed by an AI agent driving tNavigator through the Local MCP Server.

OVERVIEW

Executive Summary

The BLACK_OIL_DEMO model — a three-phase black-oil reservoir with 15 wells and 3.7 years of production history — was history matched and forecast end-to-end by an AI agent connected to tNavigator through the Local MCP Server. The engineer supervised each step and made the final decisions; the agent ran the models and the diagnostics, and produced this report.

616	65.1 2.77 →	54% 7.3% →	12% 1.6% →	2.46 M sm ³
SIMULATIONS RUN	OBJECTIVE FUNCTION	FIELD WATER ERROR	FIELD OIL ERROR	FIELD EUR

What was done

- Static model review: grid, properties, fluids
- Baseline run and mismatch diagnosis
- 200-run sensitivity screening (20 parameters)
- Differential-evolution history match, 3 rounds
- Measured BHP integrated into the objective
- Per-well segmented DCA, 10-year forecast
- Physics cross-check against the matched model

Key findings

- Rate-only matching produced a false optimum; measured BHP exposed and corrected it
- Five regional permeability multipliers and Kv/Kh dominate the match
- Final match v754: oil error 1.6%, water 7.3%
- DCA 10-year volume 1.31 M sm³; simulation 1.09 M sm³ (17% gap from pressure support)
- Field EUR: 2.46 M sm³

“The AI agent does the work — the engineer stays the engineer.”

Study workflow

1. **Connect** — the AI agent attaches to tNavigator through the Local MCP Server, one installer component launched from the GUI.
2. **Execute** — decks, 616 simulation runs, sensitivity screening and DCA, driven through the tNavigator workflow API.
3. **Supervise** — the engineer reviews diagnostics at every step, redirects methodology, and signs off the match and forecast.
4. **Report** — the agent generates interactive dashboards and this document automatically.

SECTION 01

The Simulation Model

BLACK_OIL_DEMO is a three-phase black-oil model (oil, water and gas with dissolved gas) built on a faulted corner-point grid of $48 \times 35 \times 128$ cells, of which 58,302 (27%) are active. The reservoir sits at approximately 1,414–1,475 m TVD, with two equilibration regions separated by a threshold-pressure barrier: both share a water–oil contact at 1,475 m but carry different bubble-point pressures (63.8 and 96.5 bar), reflecting compartmentalised charge. Simulation is run in metric units from 15 May 2011, with 3.7 years of well-by-well production history.

Item	Value
Model type	Black oil, 3-phase (OIL / WATER / GAS + DISGAS), Stone II
Grid	Corner-point, $48 \times 35 \times 128 = 215,040$ cells; 58,302 active (27.1%)
Structure	Faulted anticline; 502 fault-segment capacity; pinch-outs (PINCH, MINPV)
Depth / contacts	Top $\approx 1,414$ – $1,449$ m TVD; WOC 1,475 m (both regions)
Equilibration	2 regions; datum 1,450 m @ 141 bar; $P_b = 63.8 / 96.5$ bar; THPRES barrier
Saturation functions	SWOF + SGOF, 1 family; end-point scaling (ENDSCALE)
Wells	15 vertical wells: 9 producers, 6 later converted to injection
History	15 May 2011 – 01 Feb 2015, LRAT control from measured liquid rates
Units / start	METRIC, 15 May 2011

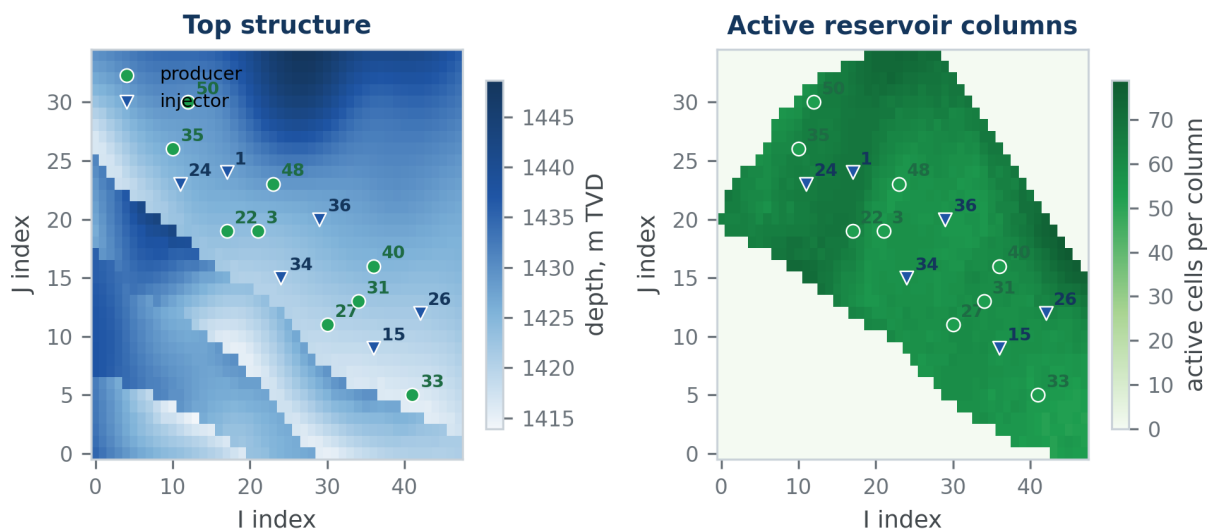


Figure 1. Top-structure depth map (left) and active reservoir columns (right), with the 15 wells posted: circles are producers, triangles are wells converted to injection. The structure dips to the north-east; the active area outlines the faulted flank blocks.

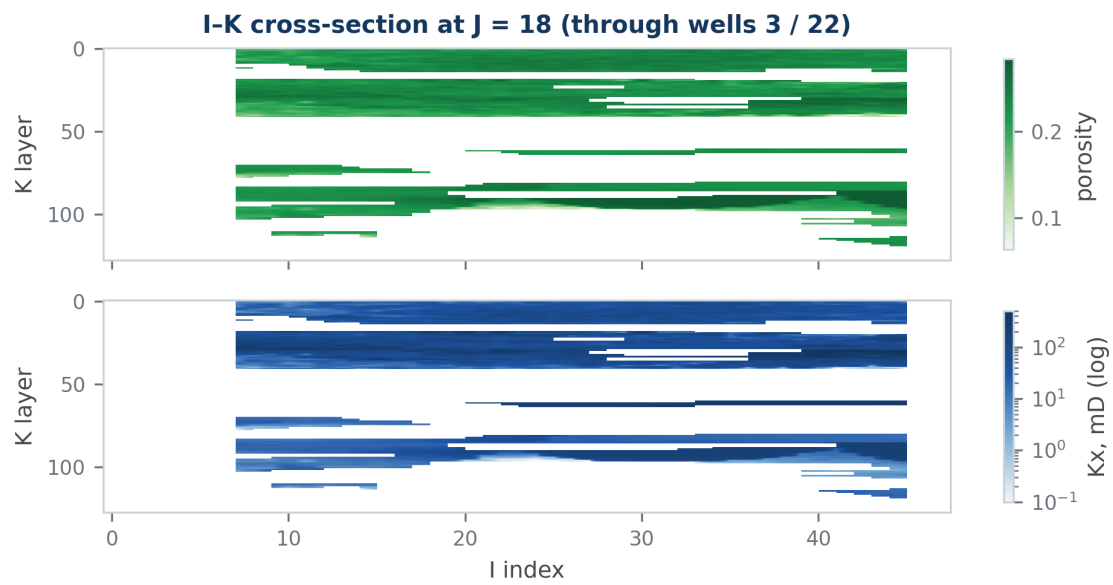


Figure 2. I-K cross-section along J=18, passing through wells 3 and 22: porosity (top) and log-permeability (bottom). The 128 thin layers preserve the vertical heterogeneity that controls water movement; low-permeability intervals separate the main flow units.

SECTION 02

Reservoir Properties

Porosity averages 22.5% across the active cells — a good-quality reservoir — while horizontal permeability spans nearly five orders of magnitude (0.006–504 mD) with a mean of 65 mD. The contrast between the P10 (5.6 mD) and P90 (182.8 mD) deciles, combined with a mean net-to-gross of 0.70, describes a strongly layered system in which thin, high-permeability streaks carry most of the flow. Vertical permeability is on average 1.5% of horizontal, limiting vertical cross-flow — one reason the history match later proved sensitive to the K_v/K_h multiplier.

Property (active cells)	Mean	P10	P90	Min	Max
Porosity, frac.	0.225	0.192	0.256	0.030	0.284
Permeability K_x , mD	64.9	5.6	182.8	0.006	503.7
Permeability K_z , mD	0.97	0.02	3.64	0.0001	10.1
Net-to-gross, frac.	0.70	0.40	1.00	0.40	1.00

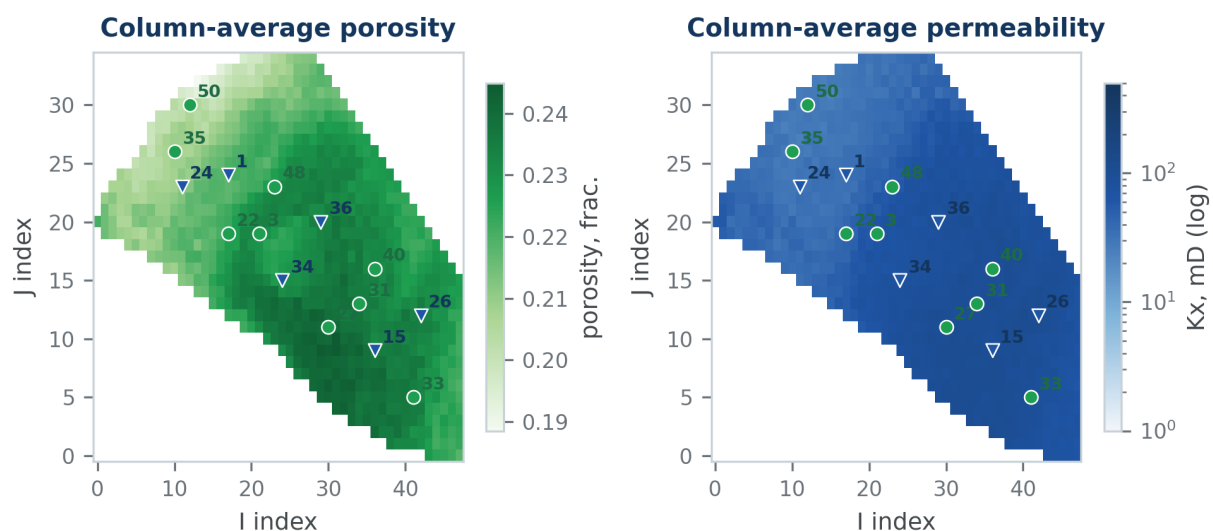


Figure 3. Column-averaged porosity (left) and log-permeability (right). The best reservoir quality tracks the crestal area where most producers are located; permeability degrades towards the flanks. Regional multipliers PERM1–PERM5 used in the history match follow these zones.

Why it matters. The permeability zonation shown here defines the five regional multipliers later used as history-match parameters. Anchoring match parameters to geological zones — rather than cell-by-cell adjustments — keeps the matched model geologically consistent and usable for forecasting.

SECTION 03

Fluids & Rock Physics

The reservoir contains a moderately volatile black oil: solution GOR rises from 27 to 126 sm^3/sm^3 along the saturated line, oil formation volume factor reaches 1.29 rm^3/sm^3 at 232 bar, and saturated viscosity falls from 2.2 to 0.75 cP with increasing pressure. Two PVT regions carry the same water and rock properties but different bubble points, matching the two equilibration compartments.

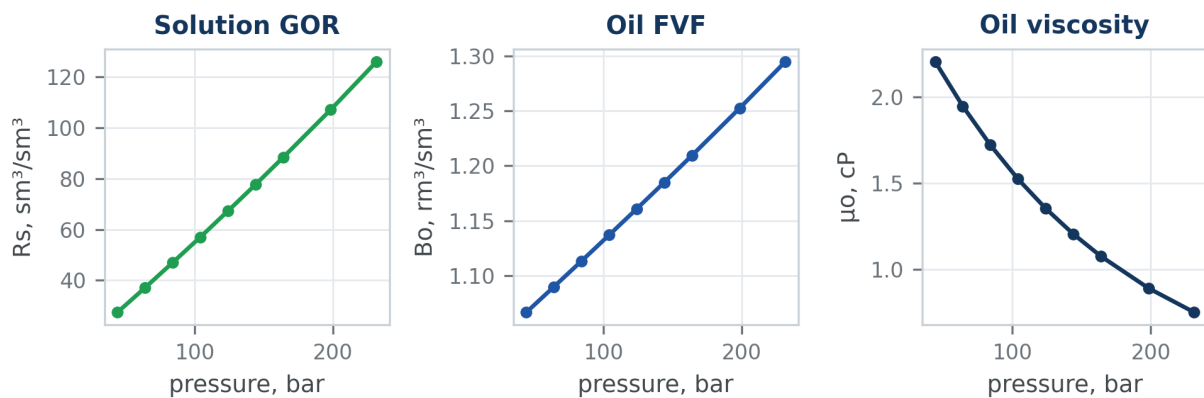


Figure 4. Saturated-oil PVT (region 1): solution GOR, oil formation volume factor and viscosity versus pressure, from the PVTO table.

Fluid / rock item	Region 1	Region 2
Surface densities (oil / water / gas), kg/m^3	832 / 1012.8 / 1.139	834 / 1012.8 / 1.139
Bubble-point pressure, bar	63.8	96.5
Water FVF @ 141 bar / viscosity, cP	1.023 / 0.41	1.023 / 0.41
Rock compressibility @ 141 bar, 1/bar	9×10^{-5}	9×10^{-5}

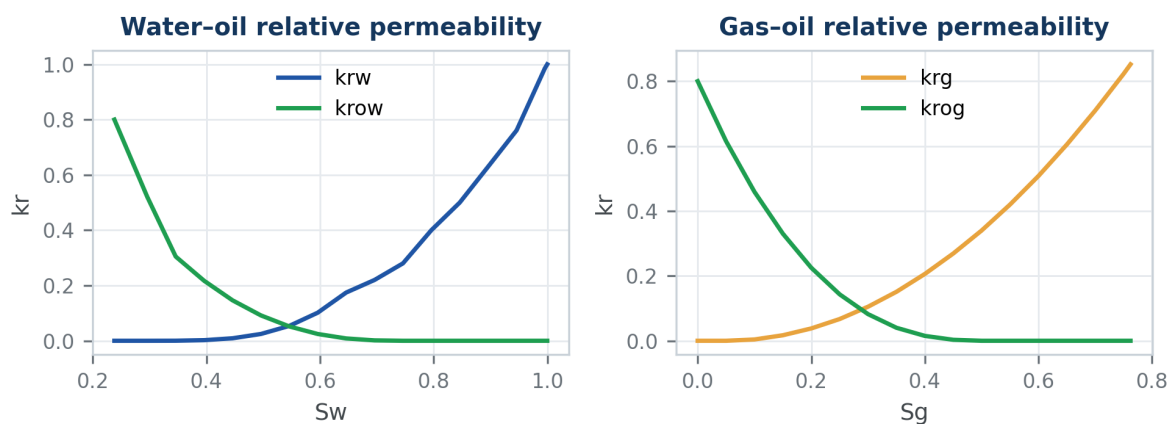


Figure 5. Relative permeability: water–oil (left) and gas–oil (right). Connate water saturation is 0.238; residual oil to water is reached at $S_w=0.746$. These curves — particularly the water-curve maximum — were among the history-match parameters (KRWM, KROM).

SECTION 04

Wells & Production History

Fifteen vertical wells were drilled between June 2011 and mid-2012, all starting as oil producers under liquid-rate control reproducing the measured history. As water cut rose, six wells — 1, 15, 24, 26, 34 and 36 — were converted to water injection for pressure support. Field oil rate peaked at roughly 1,530 sm³/day in late 2012 and declined to about 260 sm³/day by February 2015, while field water cut climbed to approximately 0.45.

Well	I, J	Status 2015	Well	I, J	Status 2015
3	22, 20	producer	1	18, 25	injector
22	18, 20	producer	15	37, 10	injector
27	31, 12	producer	24	12, 24	injector
31	35, 14	producer	26	43, 13	injector
33	42, 6	producer	34	25, 16	injector
35	11, 27	producer	36	30, 21	injector
40	37, 17	producer			
48	24, 24	producer			
50	13, 31	producer			

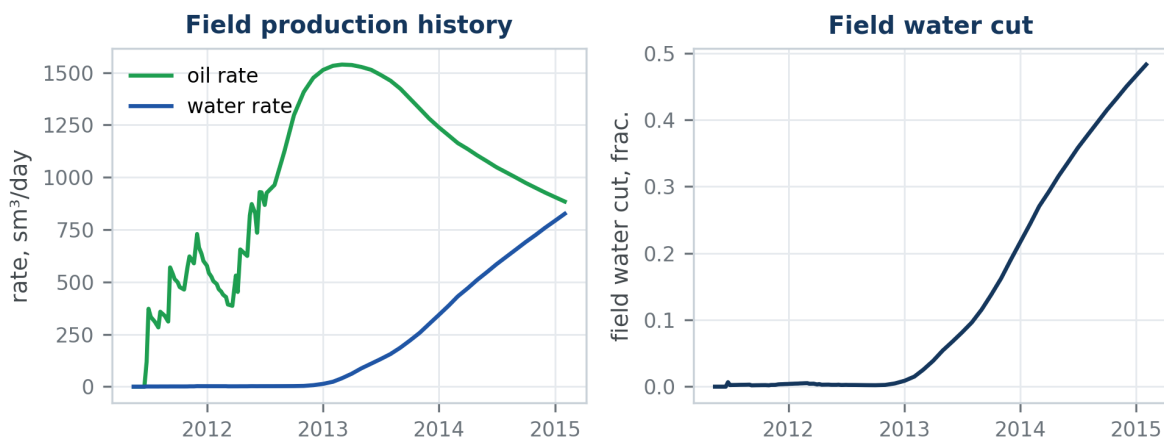


Figure 6. Field production history: oil and water rates (left) and field water cut (right), 15 May 2011 – 01 Feb 2015. The staged ramp-up reflects the drilling sequence; the water-cut trend is the primary signal the history match must reproduce.

Measured data available. Well-by-well liquid rates (LRAT history control), field oil and water volumes, and bottom-hole pressure gauge surveys in wells 3 and 40 — seven and five measurements respectively. The BHP data, although sparse, proved decisive in Section 05.

SECTION 05

Assisted History Matching

Baseline diagnosis. The unmatched model over-predicted cumulative water production by 54% and under-predicted oil by 12% (objective function 65.1). Water breakthrough timing was wrong in most wells, and base-case bottom-hole pressures drifted away from the measured gauges — clear evidence that both permeability levels and water mobility needed revision.

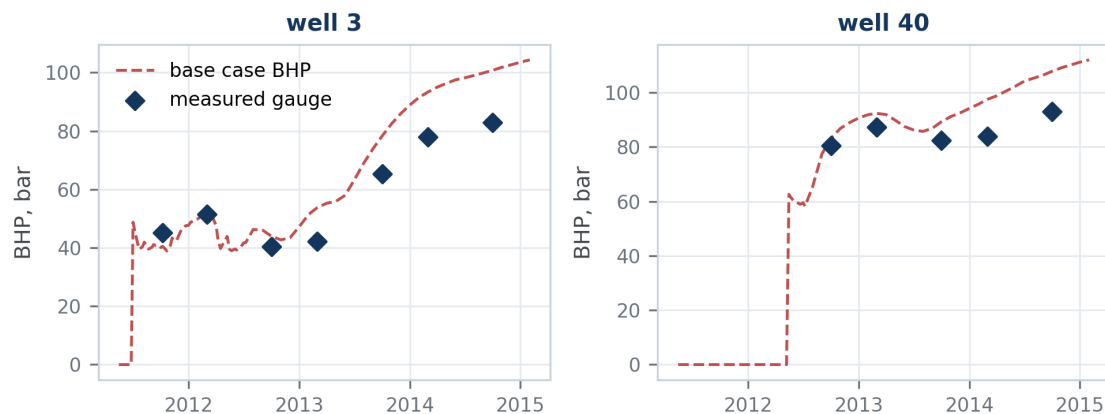


Figure 7. Base-case bottom-hole pressure against measured gauge surveys in wells 3 and 40. A rate-only match can look convincing while pressures drift away from the data — exactly what happened in round 1.

Methodology. The match proceeded in three rounds, each proposed by the agent and approved by the engineer:

1. **Rates only** — differential evolution over 9 regional parameters (five permeability-zone multipliers, K_v/K_h , water and oil rel-perm end points, a mixing coefficient). The objective fell from 65.1 to 39.0 — a seemingly good match that the measured BHP exposed as a false optimum.
2. **Rates + BHP** — the objective was extended with the well 3 and 40 gauge data and per-well productivity multipliers; the engineer rejected a proposed artificial skin adjustment in favour of a 200-run sensitivity screening of 20 candidate parameters.
3. **Local refinement** — a focused scan around well 35 (local water rel-perm and corridor multipliers) closed the remaining local mismatch and converged on variant v754.

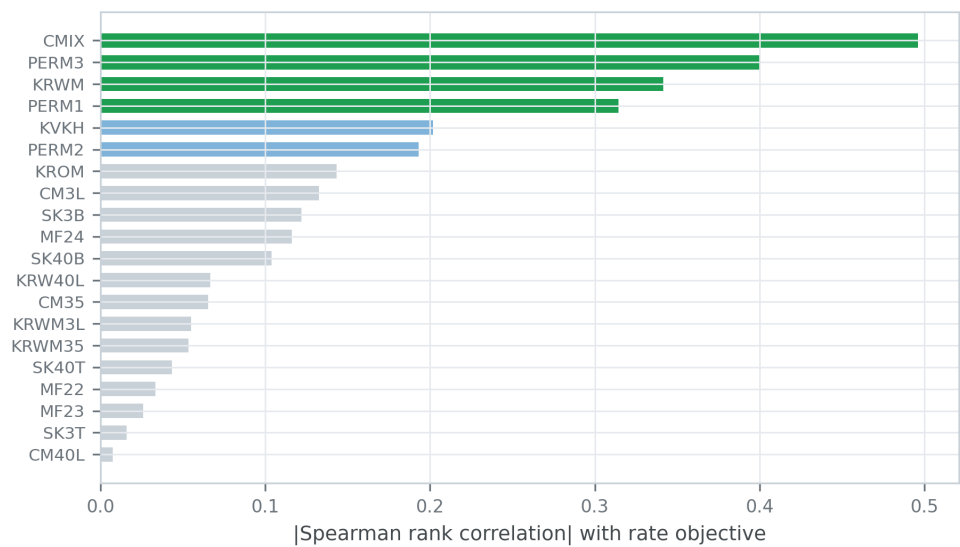


Figure 8. Sensitivity screening: rank correlation of each of the 20 candidate parameters with the rate objective across 116 Latin-hypercube runs. Regional permeability multipliers, Kv/Kh and the water rel-perm end point dominate; per-well skins have secondary influence.

Convergence and matched parameters.

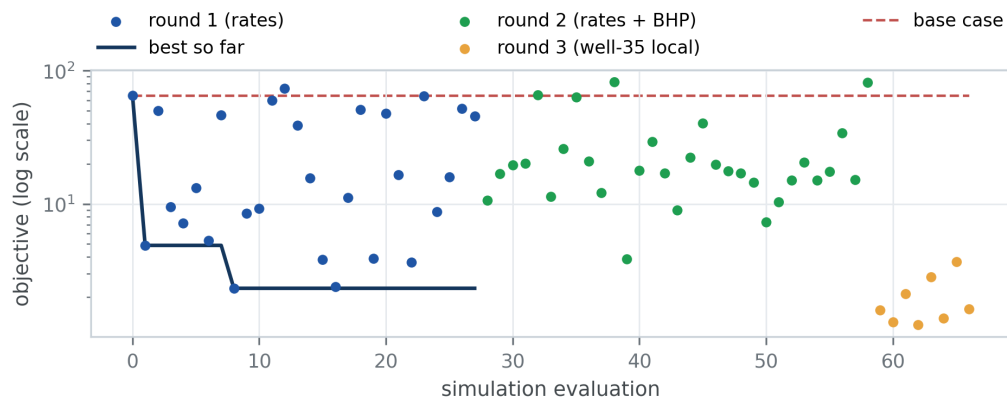


Figure 9. Objective function per simulation evaluation across the three rounds (log scale). Round 2 restarts higher because the objective was re-based to include BHP misfit; the base case is the dashed line.

Parameter	Meaning	Base	Matched (v754)
PERM1	Permeability multiplier, zone 1	1.0	1.37
PERM2	Permeability multiplier, zone 2	1.0	0.54
PERM3	Permeability multiplier, zone 3	1.0	2.09
KVKH	Vertical-to-horizontal ratio multiplier	1.0	1.50
KRWM	Water rel-perm end-point multiplier	1.0	0.80
CMIX	Corridor mixing coefficient	0.0	0.41
KROM	Oil rel-perm end-point multiplier	1.0	1.23
PI3 / PI40	Productivity multipliers, wells 3 / 40	1.0	0.55 / 0.45
KRWM35 / CM35	Local water rel-perm / corridor, well 35	1.0 / 0.0	1.20 / 0.40

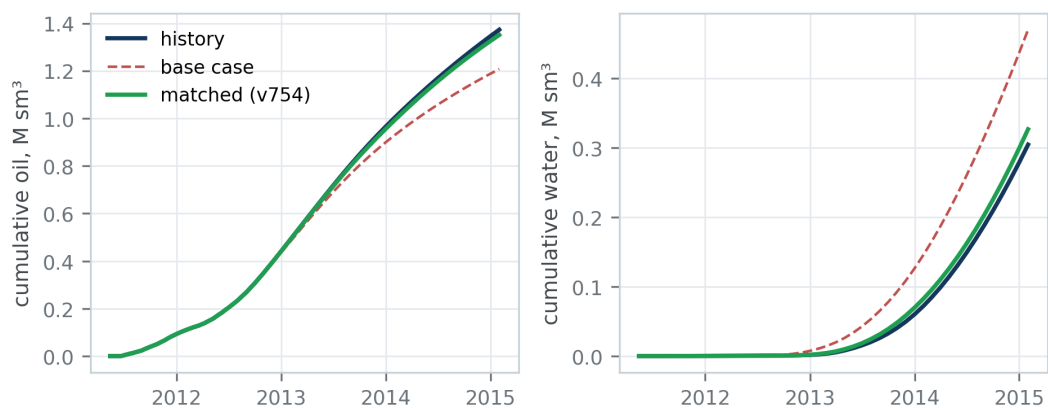


Figure 10. Cumulative field oil and water: history, base case and the selected match v754.

Match quality.

Indicator	Before	After
Objective function	65.1	2.77
Cumulative field water error	54%	7.3%
Cumulative field oil error	12%	1.6%
BHP gauge RMSE (well 3 / well 40)	—	23% / 13%
Selected variant	—	v754

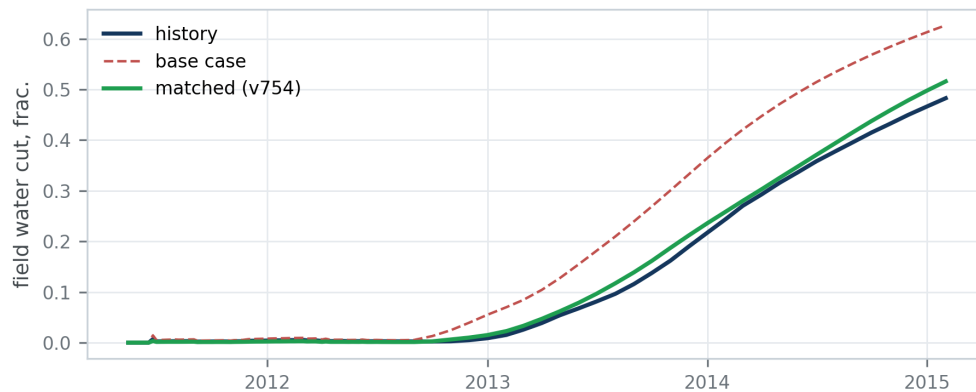


Figure 11. Field water cut: the matched model reproduces both breakthrough timing and the late-time trend that the base case missed entirely.

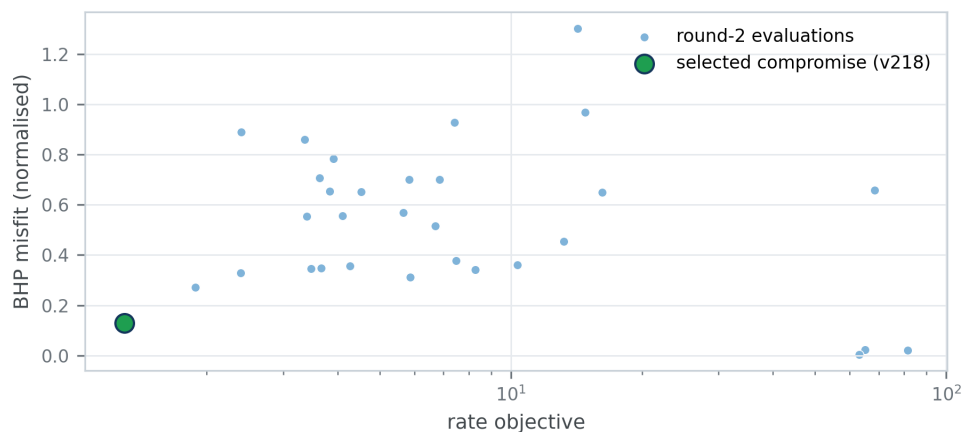


Figure 12. The rate-versus-pressure trade-off across round-2 evaluations. No single variant minimises both objectives simultaneously; the engineer selected the compromise variant on the Pareto front rather than letting the optimiser choose.

Engineering control. The agent proposed each step; the engineer rejected an artificial skin adjustment, demanded the sensitivity-first methodology and made the final Pareto trade-off. Every iteration loop took minutes, not days.

SECTION 06

Decline-Curve Analysis & Forecast

Per-well decline-curve analysis was performed on the nine active producers, with production history segmented at operating-regime changes (one Arps fit per decline cycle) and the final segment selected by AICc across exponential, harmonic and hyperbolic forms. The six injectors were excluded. Fits achieve r^2 of 0.96–1.00 on the final decline segments.

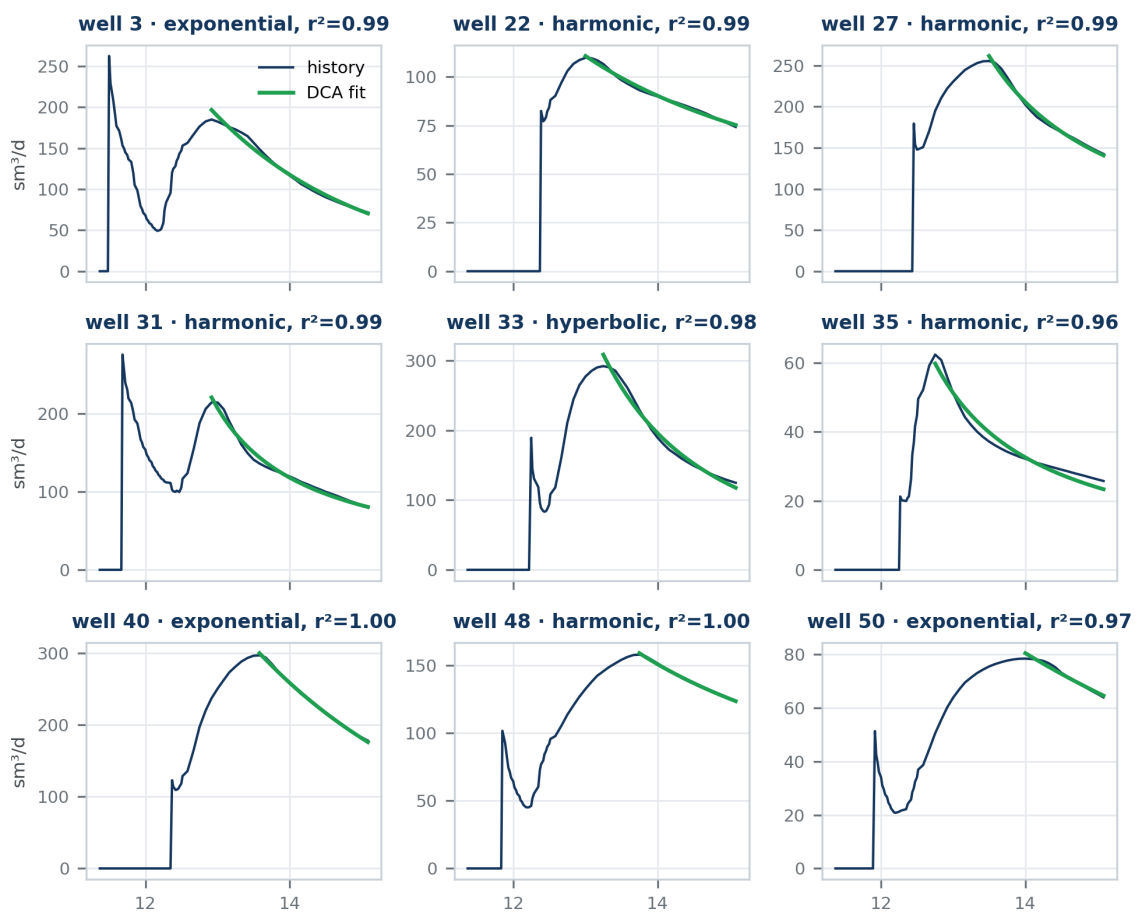


Figure 13. Per-well oil-rate history (navy) and the final-segment Arps fit (green) for the nine forecast wells. Multi-segment fitting isolates the current decline regime from earlier constrained-rate periods.

Fitted decline parameters.

Well	Model	q_i , sm^3/d	D_i , %/yr	b	r^2
3	exponential	197	47	0.00	0.987
22	harmonic	111	23	1.00	0.992
27	harmonic	262	54	1.00	0.991
31	harmonic	221	80	1.00	0.986
33	hyperbolic	309	68	0.52	0.985
35	harmonic	60	67	1.00	0.959
40	exponential	300	35	0.00	0.999
48	harmonic	159	21	1.00	0.998
50	exponential	80	20	0.00	0.970

10-year forecast and physics cross-check. The DCA forecast was run to February 2025 under an economic limit of 1 sm^3/day per well and compared with the matched-model simulation forecast as a feasibility check.

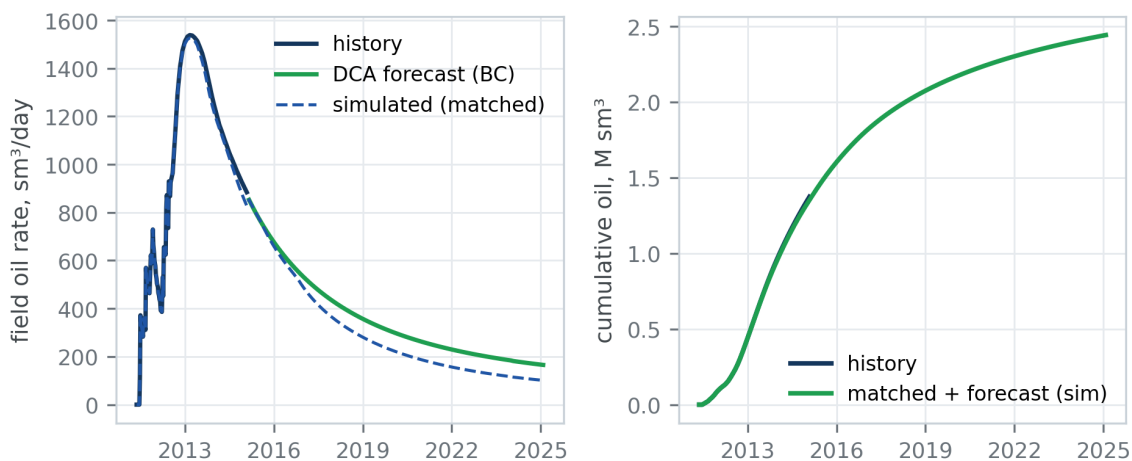


Figure 14. Field oil rate and cumulative production: history, DCA forecast under boundary conditions, and the matched-model simulation forecast.

Forecast indicator	Value
Wells forecast	9 / 15
DCA 10-year volume	1.31 M sm^3
Simulated 10-year volume	1.09 M sm^3
DCA vs simulation gap	17%
Field EUR (produced + forecast)	2.46 M sm^3

Interpretation. The 17% gap between decline-curve and simulation forecasts is driven by pressure support in the matched model that empirical declines cannot see — a reminder to anchor empirical forecasts to physics before reserves are booked. The harmonic declines fitted to several wells implicitly assume continued injection support; if injection ceases, the simulation forecast is the more reliable envelope.

SECTION 07

Conclusions & Recommendations

Conclusions

1. The BLACK_OIL_DEMO model was matched to 3.7 years of history to within 1.6% on oil and 7.3% on water, with bottom-hole pressures honoured in the two gauged wells.
2. Sparse pressure data materially changed the answer: the rate-only optimum was wrong, and two wells' worth of gauge surveys were sufficient to expose and correct it.
3. A geologically anchored parameterisation — five permeability zones, Kv/Kh and rel-perm end points — reached convergence in 616 runs without cell-level modifications.
4. DCA and simulation forecasts agree within 17%; the difference is explained by injection pressure support and bounds the forecast uncertainty.

Recommendations

1. Acquire regular BHP surveys in at least two additional producers; sparse pressure data proved decisive for match uniqueness.
2. Re-run the sensitivity screening after each new six months of history; parameter influence shifts as water cut develops.
3. Use the matched model (not standalone DCA) for decisions that depend on injection strategy; use DCA for quick-look screening.
4. Repeat this workflow quarterly — at roughly one hour of compute per full campaign, the marginal cost of staying matched is small.

“Two pressure gauges and 616 fast simulations were worth more than a perfect-looking rate match.”

SECTION 08

How This Report Was Produced

Every number, table and figure in this document was generated from the study data — the simulation decks, the optimisation campaign logs, the sensitivity study and the decline-curve fits — by an AI agent working through the tNavigator Local MCP Server. The agent parsed the grid and property arrays directly from the model, extracted the campaign results, rendered the figures in the corporate style, and compiled the document; the engineer reviewed and approved the content.

Component	Detail
Simulator	tNavigator 26.2, black-oil kernel
AI connection	Local MCP Server (installer component, launched from the GUI)
Licenses used	1 × APIS, 1 × APIC, 1 × AI
Campaign size	616 simulation runs: 3 DE rounds + 200-run sensitivity + local scans
Compute	Single workstation; about 1 hour wall-clock for the full campaign
Deliverables	Matched model, forecast decks, interactive dashboards, this report

Explore the live results

The interactive dashboards behind Sections 05 and 06 are published online:

- History match: https://lanetszb.github.io/tnav-ai-webinar/dashboards/hm_dashboard.html
- DCA forecast: https://lanetszb.github.io/tnav-ai-webinar/dashboards/dca_dashboard.html

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